

Alternative Power for C-Star at High Latitude

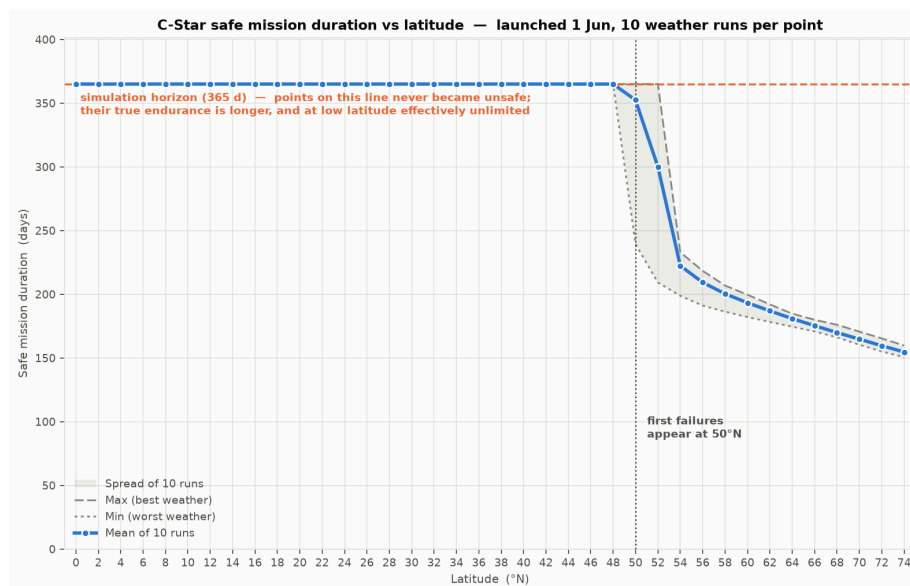
Recommendation: fit a micro wind turbine. Wave energy falls short by ~20x and should be closed out; propeller regeneration is self-limiting and cannot close the gap; a larger battery costs more mass than the vehicle can carry.

1. Sizing the problem

I modelled the baseline vehicle before looking at any harvester, because "would this help?" has no answer until you know how large the deficit is and *when* it occurs. The model runs an hourly energy balance over a year: solar geometry (so day length and polar night emerge naturally), Kasten–Young air mass, the cosine loss of a low sun onto a deck panel, persistent stochastic cloud, sail shading, and a battery with temperature-derated capacity, a sub-zero charge inhibit and ageing. All assumptions are labelled A1 – A25 inline in `cstar_power_model.py` with a note on firming each up.

Safe mission duration means time from launch until the battery first falls below 20 % of nominal — not failure, but the point at which no reserve remains for a storm or a bad weather run. All runs assume a **1 June launch**: it exercises a full seasonal cycle, it is the most favourable date so every failure is a genuine capability limit, and a vehicle reaching spring with charge in hand returns to surplus conditions and can then run indefinitely.

The problem is darkness, not energy. At 65 °N the array generates 15.0 kWh a year against an 8.8 kWh load — a 1.7x surplus — and the vehicle still dies, because that energy arrives in the wrong months. December generation is 0.1 Wh/day against 24 Wh/day consumed — one-thousandth of July. Three effects compound, all from low sun: day length collapses; the flat deck panel suffers a cosine loss (at a 1.9° sun, 97 % of available light is lost to geometry alone); and the beam crosses ten atmospheres instead of one. This is a **seasonal storage problem**, so a useful solution must deliver *in the dark months specifically*, not on annual average.



Safe mission duration vs latitude, solar only

Sweeping latitude at 2° intervals with ten weather realisations each: **every run survives the full year up to 48 °N**; 50–52 °N is a knife edge where variance is two orders of magnitude higher than elsewhere and the mean is meaningless; **from 54 °N upward every run fails**. Above 54 °N the spread between best and worst run is only 9–34 days, so endurance is set by orbital geometry rather than weather luck — predictable, but with no good year to hope for.

The sweep stops at 74 °N: beyond ~75 °N the Arctic basin is ice-covered, a 1 m sailing hull cannot operate in ice, and the model has no ice physics. Ice also damps the wave field, which matters below.

The requirement. Bisecting on constant added generation until every blackout disappears gives **0.25 W at 65 °N and 0.48 W at 70 °N** — 25–50 % of the total vehicle load. That is the target.

The goal. C-Star's value is long-duration autonomy, so the bar is binary rather than incremental: a solution must carry the vehicle **through the winter**, because surviving one winter implies surviving any number. Extending endurance from 193 to 250 days does not solve the problem; it defers it and still ends with a dead vehicle. Options are judged on that test alone.

C-Star also operates in driven mode under propeller, and an operator will only use that capability if they can trust the vehicle to recharge — so a solution that merely scrapes through winter is worth much less than one restoring usable margin.

2. The four options

A larger battery — fails on mass. Below the survival threshold, capacity buys endurance *linearly* at 3.9 days per kg ($R^2 = 0.9996$), so there is no efficiency argument against it. But the pack needed to clear winter scales badly: **3640 Wh at 60 °N (+21 kg, 53 % of vehicle mass), 4680 Wh at 65 °N (+31 kg, 77 %), 6500 Wh at 70 °N (+47 kg, 118 %)**. At 70 °N the cells alone outweigh the complete vehicle — not a modification but a different vehicle. It is the right answer for the marginal 50–52 °N band and nothing beyond, and it adds no generation: it enlarges the tank without addressing the fact that nothing fills it in December.

A wave-energy harvester — fails by ~20x. A 1 m hull is far shorter than an ocean wavelength (50–150 m), so it is a *wave follower*: it rides the surface, leaving no hull-to-water relative motion to exploit. Power can only come from an internal proof mass reacting against the hull's own acceleration — $a = \omega^2 \cdot H_s / 2$, energy per half-cycle $\sim 2 \cdot m \cdot a \cdot s$. With a 2 kg mass, 60 mm of usable stroke and 50 % PTO efficiency, mean output is **0.023 W**, ~2 % of the load, and the latitude curve is indistinguishable from solar alone. Power is proportional to stroke, and reaching 0.5 W would need a 5 kg mass moving 0.5 m inside a 1 m vehicle. **This is a scale problem, not a maturity problem** — no PTO refinement recovers a factor of twenty. Recommend closing it out.

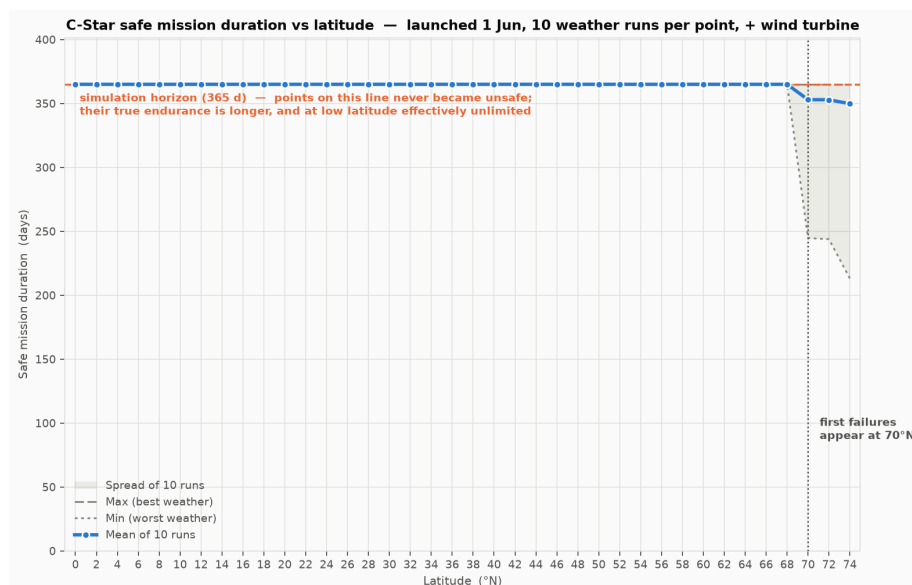
Propeller regeneration — self-limiting. Attractive because it adds no external hardware: let the existing propeller freewheel and run the motor as a generator. Water is 830x denser than air, so a small disc looks promising. But Oshen's propeller has ~2–3 cm blades, giving a ~5 cm swept disc, and power scales with diameter *squared*. At a 2 kn maximum boat speed, mean output is **0.10 W** — below the 0.25 W requirement — and the turbine costs **14 % of boat speed** (2.00 kn to 1.72 kn), computed from the equilibrium of hull and turbine drag, both of which scale as v^2 .

Crucially, **you cannot scale out of this**. The extracted power comes from the boat's kinetic energy, so a bigger disc slows the boat, and power goes as v^3 :

Disc diameter	5 cm	8 cm	10 cm	12 cm	15 cm
Mean output	0.100 W	0.154 W	0.169 W	0.173 W	0.167 W
Speed loss	14 %	28 %	36 %	43 %	51 %

Output saturates near **0.17 W** and then declines while the speed penalty grows without limit. The concept is self-limiting below requirement, extending survival only from 48 °N to 50 °N. It remains worth enabling opportunistically — it is nearly free — but it is not a solution.

A micro wind turbine — works, with margin. A 0.20 m rotor, $P = \frac{1}{2} \rho \cdot A \cdot C_p \cdot \eta \cdot v^3$, 3 m/s cut-in, 12 W rating, furling at 25 m/s. Ocean wind rises with latitude and **peaks in midwinter**, in direct antiphase with the sun — the physical basis of the concept. Wind is a persistent AR(1) daily series, shear-corrected from 10 m to a ~1 m hub (costing ~22 %), with the cubic evaluated hourly because $E[v^3]$ is roughly twice $E[v]^3$. Mean output is **2.6 W at 60 °N and 3.1 W at 70 °N** — five to ten times requirement, and 2–3x the entire vehicle load.



Safe mission duration vs latitude with a micro wind turbine

Every run survives the full year up to 68 °N, against 48 °N for solar alone. 70–74 °N becomes marginal, not because the wind fails but because a winter calm can arrive with no solar backstop.

3. Recommendation, integration and testing

Recommend the wind turbine. It is the only option meeting the requirement with margin, the only one reaching beyond ~50 °N, and the only one restoring genuine driven-mode margin rather than just averting death.

Integration is the hard part, not the power. The blockers: a rotating machine competing for masthead space with a *rotating wingsail*, which is both a geometric and an aerodynamic clash; mass and windage high up, attacking righting moment on a 40 kg hull; the vehicle's first exposed bearing, in salt spray, unattended for a year; survival through knockdown and immersion; and **icing at exactly the latitudes where it is most needed — the largest unquantified risk**. My starting concept would be a stern-mounted pylon rather than masthead, accepting lower wind speed to keep clear of the sail and keep the CG down, feeding the existing MPPT input through a rectifier.

De-risking, cheapest and most decisive first. Each test is designed to *kill* the concept early:

1. **Fix the input data (days, ~£0).** Replace synthetic cloud and wind climatology (A9 , A17) with ERA5 / NASA POWER reanalysis for the customer's real operating box and re-run. Every number here moves; nothing else should be funded first.
2. **Instrument a hull already going to sea (weeks, low cost).** IMU, GPS speed and panel current — retiring three assumptions at once: acceleration spectrum settles wave, speed distribution settles regeneration, panel current gives real shading loss.
3. **Wind-tunnel or CFD the sail-rotor interaction (~3 weeks).** The one question that decides whether the concept is integrable at all.
4. **Bench-test a candidate rotor (~2 weeks)** for its real power curve, then **cold-soak and ice test** it — the risk most likely to be found too late.
5. **Sea trial as a data logger, not a mission (~1 month).** One vehicle, everything logged, on a route where failure is recoverable. No customer mission until a full seasonal dataset exists.

Two months vs six. With **two months** the goal is a *decision*, not a prototype: steps 1–4 plus a deliberately non-flight-representative lash-up — a commercial off-the-shelf marine turbine on a stern pylon, logged in sheltered water. The deliverable is a measured power curve and a defensible answer on sail interaction. I would not attempt a sealed, flight-qualified installation.

With **six months** the first two are unchanged — those tests are on the critical path either way and their value does not scale with schedule. The extra four buy: a sealed, bearing-life-tested installation; the icing and knockdown survival campaign; a winter deployment above 60 °N validating the model against reality rather than reanalysis; and re-qualification of stability with mass aloft. The deliverable is a prototype Oshen could sell a mission on.

The distinction matters: **the two-month plan is not a compressed six-month plan**. It drops qualification, accepts a non-representative build, and buys a decision rather than a product. Attempting six months of scope in two would yield a prototype that fails for integration reasons and teaches us nothing about whether the concept was sound.

Next step. Two weeks of reanalysis data plus one instrumented deployment — a few hundred pounds — would firm up every number in this note before any hardware commitment.

4. Key assumptions

#	Assumption	Why it matters	To firm up
A9	Cloud from a latitude correlation	Sets the entire solar deficit	ERA5 / PVGIS for the real box
A17	Wind climatology and winter peak	The whole case for wind	ERA5 reanalysis
A21	Wave stroke and proof mass	Decides a 20x shortfall	IMU on a hull at sea
A24/25	Turbine thrust and hull drag	Sets the 14 % speed penalty	Tow tank
A2	Panels flat on deck	Low-sun cosine loss is punishing	CAD; check for vertical area

Two assumptions bound the study and should be confirmed with Oshen: that **no additional solar area is available**, and that **average consumption cannot be reduced below 1 W**. Since the deficit is measured in tenths of a watt, both are disproportionately powerful levers — a winter duty-cycle reduction may well be cheaper than any hardware considered here.

Supporting work: `cstar_power_model.py` , `run_mission.py` , `sweep_latitude.py` , `sweep_battery.py` .